

Sumit Pradeep Kolpe

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EDUCATION

Savitribai Phule Pune University (MMCOE, Pune, India)

Oct 2022 – June 2026

Bachelor of Engineering in Artificial Intelligence & Data Science

GPA: 9.85/10

ACADEMIC PROJECTS

Saheli: AI-Powered Women's Safety Platform [MMCOE, Pune] [[Website](#)]

Jan 2024 – Present

Team Size: 4

Role: Lead Researcher & NLP Developer

- Devised a multimodal AI system to detect distress from voice and text and trigger silent emergency alerts.
- Implemented privacy-preserving real-time location sharing and secure alert pipelines to protect user data.
- Trained and optimized NLP and classification models using Python and TensorFlow, improving detection accuracy by 20% over baseline and reducing average emergency response time to under 3 seconds.
- Awarded competitive research funding from the Government of India (AICTE) to support system development.

FitMantra: AI-Based Virtual Gym Trainer [MMCOE, Pune] [[Website](#)]

Jan 2023 – May 2024

Team Size: 7

Role: Team Leader & Computer Vision Engineer

- Engineered a computer-vision posture-correction system using OpenCV and TensorFlow to provide real-time corrective feedback.
- Integrated gamified progress tracking and community leaderboards to boost user engagement.
- Optimized pose-estimation pipelines and evaluation metrics, achieving an approximate 60% reduction in posture errors during testing.

INTERNSHIPS

International Research Intern, Avignon University, France

June 2025 – August 2025

- Led the development of a digital twin simulation for 5G/6G wireless networks to evaluate real-time behavior under DoS attacks.
- Implemented AI-based anomaly detection to strengthen network resilience and optimize defense response time.
- Explored the role of Reconfigurable Intelligent Surfaces (RIS) and ethical AI policies for secure network governance.

AI Practitioner Intern, Infosys, Pune India

May 2024 – July 2024

- Developed and deployed time-series forecasting models for demand prediction in e-commerce, boosting accuracy by 25%.
- Optimized model interpretability and explainability to align with Infosys's ethical AI guidelines.
- Collaborated with cross-functional data teams to improve forecasting pipelines and streamline business decision-making.

Data Scientist Intern, Sponsorlytix, United Kingdom

Nov 2022 – Dec 2022

- Designed and optimized a multimodal driver-attention and hazard-forecasting system using in-cabin behavior + forward-road video.
- Improved attention-detection to 92%+ F1, reduced false positives by ~33%, and enabled ~2.4s earlier risk-warning predictions by rebuilding a noisy real-world data pipeline.

TECHNICAL SKILLS

- Programming Languages:** Python (3 yrs), C++ (2 yrs), SQL (2 yrs).
- Machine Learning & AI:** TensorFlow (2 yrs), PyTorch (1.5 yrs), Scikit-learn (3 yrs), OpenCV (2 yrs).
- Data Science & Analytics:** Feature Engineering (3 yrs), Data Cleaning (3 yrs), NLP (2 yrs).
- Database Systems:** MySQL (2 yrs), PostgreSQL (1.5 yrs), MongoDB (1 yrs).
- Tools & Libraries:** Pandas, NumPy, Matplotlib, Git, Jupyter Notebook, VS Code.

CERTIFICATIONS & WORKSHOPS

- Databricks Certified Machine Learning Associate 2025
- Oracle Cloud Infrastructure 2025 Data Science Professional: Oracle University 2025
- Research Symposium on AI Governance & Security: IIT Bombay, India 2025
- Oracle Cloud Infrastructure, Generative AI Certified Professional: Oracle University 2024
- Microsoft Fabric Analytics Engineer Associate: Microsoft Learn 2024

- IBM Certified Machine Learning for Data Science 2024
- Applications of Machine Learning in Urban Studies: Indian Space Research Organization (ISRO) 2023

PUBLICATIONS

- S. P. Kolpe, "An IoT-Based Smart Fitness Solution (FitMantra)," Indian Patent Application No. 202421004780, Published Feb. 2024. (Verifiable at: [Intellectual Property India](#))
- Kolpe, S. P., J. S. Jeyanathan, and P. Nagaraj, "Smart Emergency Detection Framework for Women's Safety Using Deep Learning," Journal of Big Data (Springer, Q1), under review, 2025.
- S. P. Kolpe, A. Benslimane, "RIS-Assisted Secure 6G Communication Using Intelligent Beam Reconfiguration," Manuscript in preparation, Avignon University, France, 2025.

ACHIEVEMENTS

- National Aegis Graham Bell Award 2025, Ministry of Electronics & IT, Government of India.
- Entrepreneurship Excellence Award 2025 from ENSIN Forum & VIT Pune.
- GATE Exam 2025 - Qualified in First Attempt (Top 10% Rank among 57054 Candidates Nationwide).
- Best AI Tech Award 2024, Infosys India.
- Represented India at the International Hackathon 2024, Slovakia (Europe) - Selected among 30 global finalist teams representing 30 countries.
- International Hackathon 2024 Pune, Maharashtra, 1st Prize for AI- based Virtual Gym Trainer Project.
- PAN IIT National Hackathon 2023 - 3rd Rank for AI for Women's Safety Project.
- State-Level Hackathon 2024 - 2nd Prize for AI- based Virtual Gym Trainer Project.
- Top Team Award in Idea Pitching - Innovation, Design & Entrepreneurship Bootcamp, Ministry of Education.
- Pioneer of AI & Data Science - Departmental Honor 2024.
- Academic Topper Award - AI & Data Science (3 Consecutive Years).
- Vidyadhan Merit Scholarship - Awarded by the Sarojini Damodaran Foundation for sustained academic excellence; supported education and living expenses from Class XI onward.
- Cadence "Joining Hands" Merit Scholarship (Fully Funded Undergraduate Education) - Awarded for academic excellence and leadership in technology.

LEADERSHIP ROLES

Innovation Coordinator – Institute Innovation Council (IIC), Government of India 2025 – Present

- Led design thinking and innovation initiatives, mentoring 130+ student teams in translating technical ideas into socially responsible solutions across safety, healthcare, and sustainability.
- Conducted structured innovation workshops to promote ethics-driven design and responsible AI thinking.

President – NEURA: AI & Tech Club – MMCOE, Pune 2024 – 2025

- Led a community of 100+ students, fostering research and innovation in AI and Data Science.
- Organized 3 major campus events, enhancing interdisciplinary collaboration and technical engagement.

COMMUNITY INVOLVEMENT

National Service Scheme (NSS) Volunteer – MMCOE, Pune 2023 – 2025

- Organized digital literacy and education drives across rural schools, impacting over 300 students.
- Assisted in health awareness campaigns and environmental clean-up programs benefiting 500+ community members.

Orphanage & Elder Care Volunteering – Pune Region 2023 – 2024

- Supported education and recreational programs at local orphanages, benefiting 50+ children.
- Initiated weekend learning sessions in basic computer literacy and STEM awareness for underprivileged youth.